

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Simulate cross correlation and autocorrelation on discrete time signals.	
Lab Experiment : 3	Date: 30/07/2026	Enrolment No:92400133050

Aim : Simulate cross correlation and autocorrelation on discrete time signals.

Code :

```
import numpy as np
import matplotlib.pyplot as plt

def cross_correlation(signal1, signal2):
    # Compute the cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr

def autocorrelation(signal):
    # Compute the autocorrelation
    auto_corr = np.correlate(signal, signal, mode='full')
    return auto_corr

# Define the discrete-time signals
signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10])

# Compute the cross-correlation
cross_corr = cross_correlation(signal1, signal2)

# Compute the autocorrelation
auto_corr = autocorrelation(signal1)

# Plot the cross-correlation and autocorrelation signals
plt.figure(figsize=(10, 6))
```

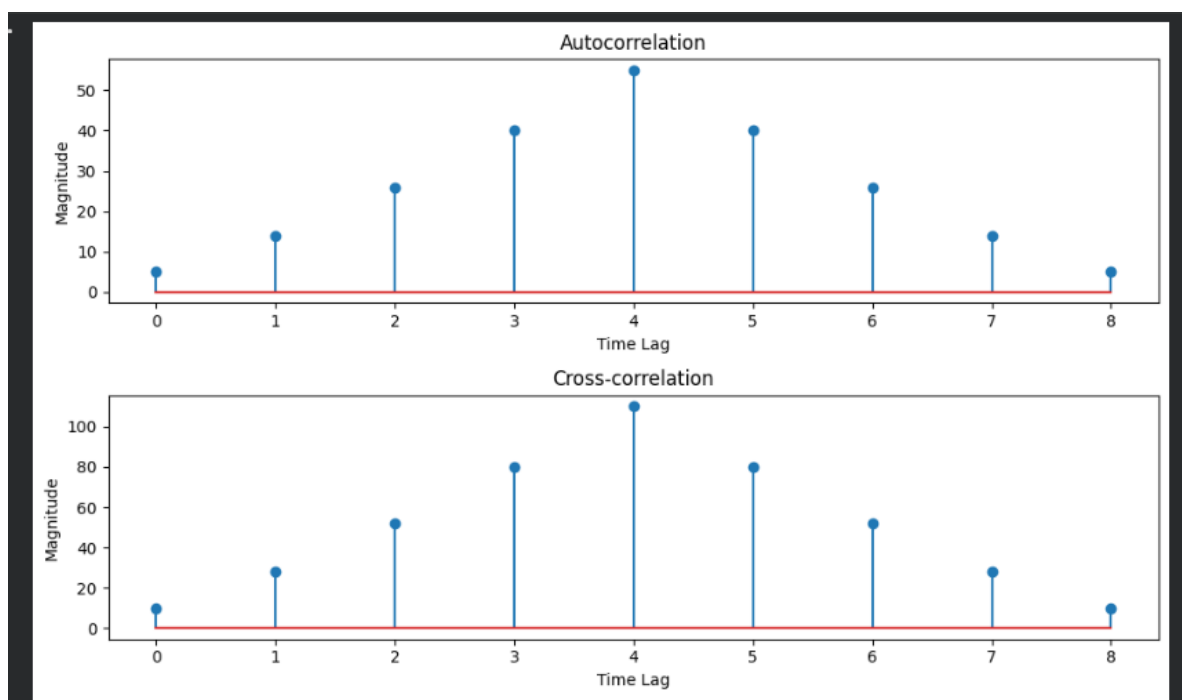
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```
plt.subplot(2, 1, 1)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
```

```
plt.subplot(2, 1, 2)
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
```

```
plt.tight_layout()
plt.show()
```

Output :



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Conclusion : In this experiment We learned how to find autocorrelation and cross-correlation of discrete-time signals using Python. The autocorrelation result showed the similarity of a signal with itself at different time lags, producing a maximum value at zero lag. The cross-correlation result measured the similarity between two different signals and demonstrated their relationship across various time shifts.



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Calculation :

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Experiment 3

$x(n) = [1, 2, 3, 4, 5]$ $n = 0, 1, 2, 3, 4$
 $y(n) = [2, 4, 6, 8, 10]$ $n = 0, 1, 2, 3, 4$

Auto Correlation

$R_{xx}[k] = \sum_{m=-\infty}^{\infty} x[m]x[m-k]$

k	Calculation	$R_{xx}(k)$
-4	1×5	5
-3	$1 \times 4 + 2 \times 5$	14
-2	$1 \times 3 + 2 \times 4 + 3 \times 5$	26
-1	$1 \times 2 + 2 \times 3 + 3 \times 4 + 4 \times 5$	40
0	$1^2 + 2^2 + 3^2 + 4^2 + 5^2$	55
1	$2 \times 1 + 3 \times 2 + 4 \times 3 + 5 \times 4$	40
2	$3 \times 1 + 4 \times 2 + 5 \times 3$	26
3	$4 \times 1 + 5 \times 2$	14
4	5×1	5

Auto Correlation: $[5, 14, 26, 40, 55, 40, 26, 14, 5]$



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Cross Correlation

$$R_{xy}[k] = \sum_{n=-\infty}^{\infty} x[n]y[n-k]$$

K	Shifted Signal $x(n-k)$	Overlapping Sym Calculation	$R_{xy}[k]$
-4	[0, 0, 0, 0, 2, 4, 6, 8, 10]	1×10	10
-3	[0, 0, 0, 2, 4, 6, 8, 10, 0]	$1 \times 8 + 2 \times 10$	28
-2	[0, 0, 2, 4, 6, 8, 10, 0, 0]	$1 \times 6 + 2 \times 8 + 3 \times 10$	52
-1	[0, 2, 4, 6, 8, 10, 0, 0, 0]	$1 \times 4 + 2 \times 6 + 3 \times 8 + 4 \times 10$	80
0	[2, 4, 6, 8, 10, 0, 0, 0, 0]	$1 \times 2 + 2 \times 4 + 3 \times 6 + 4 \times 8 + 5 \times 10$	110
1	[0, 2, 4, 6, 8, 10, 0, 0, 0]	$2 \times 2 + 3 \times 4 + 4 \times 6 + 5 \times 8$	80
2	[0, 0, 2, 4, 6, 8, 10, 0, 0]	$3 \times 2 + 4 \times 4 + 5 \times 6$	52
3	[0, 0, 0, 2, 4, 6, 8, 10, 0]	$4 \times 2 + 5 \times 4$	28
4	[0, 0, 0, 0, 2, 4, 6, 8, 10]	5×2	10

Cross Correlation: [10, 28, 52, 80, 110, 80, 52, 28, 10]